

Intestinal Goblet Cells Play a Protective Role Against GVHD Via a Lypd8-Dependent Manner after Allogeneic Hematopoietic Stem Cell Transplantation

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Abstract

[Introduction] Emerging evidences suggest that perturbations in the gut microbiota are associated with graft-versus-host disease (GVHD), and dominance of *Enterobacteriaceae* is related to poor prognosis after allogeneic hematopoietic stem cell transplantation (SCT) (Taur Y, *Blood*. 2014; 124:1174-1182). We recently reported that degree of goblet-cell loss was significantly correlated with poor prognosis in 90 patients who underwent SCT in our institute (Ara, et al. 2018 Tandem BMT meeting #220). Goblet cells play a critical role in forming the mucus layer that constitutes not only physical but also chemical barrier, by retaining antimicrobial peptides, against invading microbes from gut lumen. In the current study, we explored the mechanism by which goblet cells protect recipients against GVHD, especially focusing on the role of antimicrobial peptide Lypd8 that is produced by colon epithelial cells and specifically suppresses motilities and biotranslocation of flagellated bacteria including harmful *Enterobacteriaceae*.

[Methods] Mice were lethally irradiated and injected with 5×10^6 bone marrow cells and 7.5×10^6 splenocytes from allogeneic or syngeneic donors on day 0. Recipient mice were intraperitoneally injected with 0.3 mg recombinant mouse IL-25 (rmIL-25) or vehicle from day -6 to 0. **[Results]** In the B6 → B6D2F1 model, goblet cells in the colon were significantly decreased in association with severe GVHD (Figure A). Fluorescent in situ hybridization (FISH) using the universal bacterial probe EUB338

showed bacterial translocation to the colonic mucosa after SCT. Quantitative PCR targeting bacterial 16S rRNA confirmed that bacterial load in the lamina propria was significantly increased in allogeneic mice compared to syngeneic controls and naïve mice (Figure B). Immunofluorescent staining showed Lypd8 at the border of the inner mucus layer and colonic epithelial cells was reduced in allogeneic mice. To evaluate role of Lypd8 in GVHD, lethally irradiated B6-*Lypd8*^{-/-} mice and wild type (WT) B6 controls were transplanted from BALB/c mice. Both FISH and quantitative PCR showed an increased bacterial translocation into the colonic mucosa in Lypd8-deficient recipients compared to WT recipients (Figure C), indicating that Lypd8 plays a protective role against bacterial translocation into the colonic mucosa. In association with enhanced bacterial translocation, significantly more donor T cells were infiltrated into the gut and liver in *Lypd8*^{-/-} recipients compared to WT controls (Figure D). Strikingly, GVHD was significantly more severe with shorter survival in B6-*Lypd8*^{-/-} mice compared to WT recipients (Figure E). GVHD exacerbation in *Lypd8*^{-/-} mice was reproduced when *Lypd8*^{-/-} recipients were co-housed with WT recipients for 4 weeks before SCT, excluding the virulent microbiota in *Lypd8*^{-/-} mice as the potential mechanism of GVHD exacerbation in these mice. Finally, we tested if the goblet-cell protection could attenuate GVHD. Pre-transplant administration of IL-25 mitigated goblet-cell loss and bacterial translocation, reduced plasma levels of IFN- γ and IL-6, and ameliorated GVHD mortality. Protective effects of pre-transplant IL-25 was abrogated when *Lypd8*^{-/-} mice were used as recipients, suggesting that goblet cells protect recipients against GVHD via a Lypd8-dependent manner. **[Conclusion]** Our results demonstrated that goblet cells suppress bacterial translocation into the colon mucosa and play a protective role against GVHD via a Lypd8-dependent manner. Since increase in flagellated bacteria in the gut could be associated with GVHD exacerbation, goblet cells and Lypd8 could be potentially prophylactic and therapeutic targets for GVHD.

Figures: (A) The amount of goblet cells in the colon of allogeneic or syngeneic recipients, or naïve mice are shown. **(B)** Colon samples were harvested from syngeneic and allogeneic recipients on day +7 after SCT. DNA was extracted from colon samples after removing epithelial cells by incubating with EDTA and subjected to quantitative PCR with 16S rRNA specific primers. **(C-E)** WT or *Lypd8*^{-/-} B6 mice were lethally irradiated and transplanted from allogeneic BALB/c mice. Bacterial load in the colon lamina propria **(C)** and absolute numbers of donor T cells in the liver and colon **(D)** on day +5, and survival curves **(E)** after SCT are shown. *, p<0.05, **, p<0.01, ***, p<0.005.

Disclosures

No relevant conflicts of interest to declare.

Author notes

* Asterisk with author names denotes non-ASH members.

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